

Docker Compose Setup for Spring Boot and MongoDB

Overview:

This docker-compose.yml file defines a multi-container setup for a Spring Boot Microservice application that interacts with a MongoDB database. It uses Docker Compose v3.8, supporting advanced networking and environment management.

Services:

- MongoDB Service (employeedb)
 - * Image: mongo:latest
 - * Container Name: employeedb
 - * Port Mapping: 27017 -> 27017
 - * Auto-creation of database when first data is inserted.
- Spring Boot Application (employeeservice)
 - * Image: employee-service:1.1
 - * Build Context: ../../..
 - * Port Mapping: 9090 -> 8080
 - * Environment Variable: SPRING_DATA_MONGODB_URI
 - * Connected to employeenet

Network:

A custom bridge network named 'employeenet' is used for secure internal communication and service discovery.

Spring Boot Configuration:

server:

port: 8080

spring:

data:

mongodb:

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uri: \${SPRING_DATA_MONGODB_URI}

Usage:

- Stop all running containers:
docker-compose down
- Build and start fresh containers:
docker-compose up --build

Benefits:

- Single command setup for development
- Clean separation of services
- Flexible configuration
- Secure, isolated Docker network
- Portable across environments

Summary:

This setup creates a lightweight, scalable, and manageable microservice ecosystem with Spring Boot and MongoDB working seamlessly together over Docker Compose.

Architecture Diagram:

Docker Network: employeenet

[employeedb (MongoDB)]

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[employeeservice (Spring Boot)]